

Event Management System

Project Type: Web-Based Application

Development Stack: MERN Stack

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1. Project Selection

The selected project for this semester is the **Event Management System**. The system will be developed as a web-based application using the MERN stack. It will provide a centralized platform for creating, managing, discovering, and registering for events.

The system will help event organizers manage event information and participants while allowing users to easily find and register for available events.

2. Problem Statement

Managing events through manual records, spreadsheets, paper forms, and separate communication channels can lead to several problems. Event organizers may face difficulties in maintaining event information, tracking participants, managing registrations, and communicating event details. Users may also find it difficult to discover available events and register for them efficiently.

Therefore, a centralized **Event Management System** is needed to simplify event planning, event registration, participant management, and access to event information through a single web-based platform.

3. Project Objectives

3.1 Main Objective

To develop a web-based **Event Management System** that simplifies the creation, management, discovery, and registration of events through a centralized platform.

3.2 Specific Objectives

The system aims to:

1. Provide secure user registration and login.
2. Allow authorized users to create and manage events.
3. Allow users to browse and search for available events.
4. Allow users to register for events online.
5. Maintain event and participant records in a centralized database.
6. Provide administrators with a dashboard for monitoring events and registrations.
7. Allow event organizers to manage participant information.
8. Provide basic reports and summaries related to events and registrations.

4. Stakeholders

The main stakeholders of the Event Management System are:

Stakeholder	Role / Responsibility
Administrator	Manages users, events, registrations, and overall system activities.
Event Organizer	Creates and manages events and monitors participant registrations.
Student / User	Browses events, views event details, and registers for events.
Management	Reviews event information and monitors basic event activities.

5. Project Scope

The project scope defines the features that will and will not be included in the Event Management System.

5.1 Included Features

The system will include:

- User registration
- User login and logout
- User dashboard
- Role-based access
- Event creation
- Event viewing
- Event updating
- Event deletion
- Event search and filtering
- Event registration
- Participant management
- Event registration tracking
- Event details management
- Basic reports and summaries
- Administrator dashboard

5.2 Excluded Features

The following features will not be included in the current version:

- Dedicated Android or iOS mobile application
- Online payment gateway

- AI-based event recommendations
- Advanced third-party integrations
- Live video streaming
- Advanced predictive analytics
- Biometric attendance
- Complex QR-based ticketing system

These features may be considered for future versions of the system.

6. Functional Requirements

Functional requirements describe **what the system should do**.

ID	Functional Requirement	Description
FR-01	User Registration	The system shall allow users to create an account.
FR-02	User Login	The system shall allow registered users to securely log in.
FR-03	User Logout	The system shall allow users to securely log out.
FR-04	Role Management	The system shall provide appropriate access based on user roles.
FR-05	Event Creation	Authorized users shall be able to create new events.
FR-06	Event Update	Authorized users shall be able to update event information.
FR-07	Event Deletion	Authorized users shall be able to delete events.
FR-08	Event Viewing	Users shall be able to view event details.
FR-09	Event Search	Users shall be able to search for available events.
FR-10	Event Filtering	Users shall be able to filter events based on relevant information such as date or category.
FR-11	Event Registration	Users shall be able to register for available events.
FR-12	Participant Management	Organizers shall be able to view and manage participant records.
FR-13	Registration Tracking	The system shall maintain event registration information.

FR-14	Dashboard	The system shall provide dashboards containing relevant event information.
FR-15	Reports	Authorized users shall be able to view basic event and registration summaries.

7. Non-Functional Requirements

Non-functional requirements describe **how well the system should perform**.

Category	Requirement	Description
Security	Data protection	User information and credentials should be protected from unauthorized access.
Performance	Fast response	Pages and common operations should respond within a reasonable time.
Usability	Easy navigation	The interface should be simple, clear, and easy to understand.
Reliability	Consistent operation	The system should operate correctly and minimize errors.
Scalability	Future expansion	The system should support additional users, events, and features in the future.
Maintainability	Organized code	The application should use a clear structure that makes future maintenance easier.
Compatibility	Browser support	The system should work correctly on commonly used modern web browsers.
Availability	System accessibility	The system should be accessible during normal usage and testing periods.

8. Core Modules

The main modules of the Event Management System will be:

8.1 Authentication and User Management

This module will handle user registration, login, logout, and user roles.

8.2 Administrator Dashboard

The administrator dashboard will provide an overview of users, events, registrations, and other important system information.

8.3 Event Management

This module will allow authorized users to create, view, update, and delete events.

8.4 Event Discovery and Search

Users will be able to browse available events and search or filter events based on relevant information.

8.5 Event Registration

Users will be able to register for events through the system.

8.6 Participant Management

Event organizers will be able to view and manage users who have registered for their events.

8.7 Reports and Summaries

The system will provide basic information and summaries related to events and registrations.

9. Initial Project Summary

Project Title

Event Management System

Problem Statement

Traditional methods of managing events can make event creation, registration, communication, and participant tracking difficult and time-consuming. A centralized system is required to make these activities more organized and efficient.

Objective

To develop a web-based system that allows users and event organizers to manage events and registrations through a centralized platform.

Target Users

- Administrators
- Event Organizers
- Students / General Users
- Management

Core Modules

- User Authentication
- Administrator Dashboard
- Event Management
- Event Search and Filtering
- Event Registration
- Participant Management
- Reports and Summaries

Scope

The system will include user authentication, event management, event search, event registration, participant management, dashboards, and basic reports.

The system will not include mobile applications, online payment gateways, AI recommendations, advanced analytics, or complex third-party integrations.

Expected Outcome

The expected outcome is a functional MERN-based web application that simplifies event planning, event discovery, online registration, and participant management. The system will provide a centralized platform that improves the efficiency and organization of event-related activities.

10. Project Documentation Structure

The project documentation will be stored in the GitHub repository inside a dedicated **docs** folder.

Recommended structure:

```
Event-management-system/  
|  
├── Docs/  
|   └── Project-Planning-Document.docx  
|  
├── Client/  
|  
├── Server/  
|  
├── README.md  
|  
└── package.json
```

After completing the document, it should be added and committed to the GitHub repository.

11. Conclusion

The planning and requirement analysis phase provides a clear direction for developing the Event Management System. By identifying the problem, stakeholders, project scope, and system requirements before starting development, the project team can focus on the most important features and avoid unnecessary complexity.

The documented requirements will guide the development team throughout the semester and provide a clear foundation for designing and implementing the Event Management System.